American Computer Science League

2022-2023 ● Contest 3: Short Problems Solutions ● Intermediate Division

1. Boolean Algebra

$$\begin{aligned}\overline{A}(A + \overline{B}) + \overline{A}\,B &= \overline{A}A + \overline{A}\,\overline{B} + \overline{A}\,B \\ &= \overline{A}\,\overline{B} + \overline{A}\,B \\ &= \overline{A}(\overline{B} + B) \\ &= \overline{A}\end{aligned}$$

B. $\overline{A}$

2. Boolean Algebra

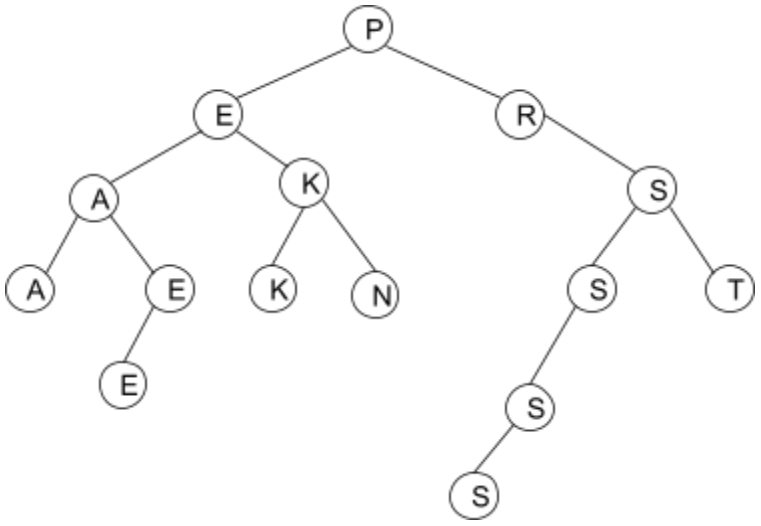
$$\begin{aligned}(\overline{A\,\overline{B} + C})\,(\overline{B\,\overline{C} + A}) &= \overline{A}\,\overline{\overline{B}}\,\overline{C}\,\overline{B}\,\overline{\overline{C}}\,\overline{A} \\ &= \overline{A}\,\overline{C}\,(\overline{A} + \overline{\overline{B}})(\overline{B} + \overline{\overline{C}}) \\ &= \overline{A}\,\overline{C}\,(\overline{A}\,\overline{B} + \overline{A}\,C + B\,\overline{B} + B\,C) \\ &= \overline{A}\,\overline{B}\,\overline{C} + \overline{A}\,C\,\overline{C} + 0 + \overline{A}\,B\,C\,\overline{C} \\ &= \overline{A}\,\overline{B}\,\overline{C}\end{aligned}$$

$\overline{A}\,\overline{B}\,\overline{C} = 1 \Rightarrow \overline{A} = 1,\ \overline{B} = 1, \overline{C} = 1$
So (0, 0, 0) is the only triple that makes the expression TRUE.

A. 1

3. Data Structures

The binary search tree for PREAKNESSSTAKES is as follows:



There are 3 nodes with only a left child: E,S, and S.

B. E, S, S

4. Data Structures The queue is constructed using FIFO as follows: R, RO, ROM, OM, M, ME, MEO, EO, O, OA, A, AN, AND, ANDJ, ANDJU, NDJU, NDJUL, DJUL, DJULI, DJULIE, JULIE, ULIE, ULIET, LIET The next item popped would be L.	D. L
5. FSAs and Regular Expressions regular expression: [b-ps-z]* [aei] [j-p]* Dances: waltz: fails at t in [j-p]* tango: fails at g in [j-p]* chacha: fails at 2nd c in [j-p]* polka: matches samba: fails at b in [j-p]* foxtrot: fails at r in [aei] charleston: fails at r in [aei] paso: fails at s in [j-p]* salsa: fails at 2nd s in [j-p]* rhumba: fails at r in [b-ps-z]* mambo: fails at b in [j-p]* quickstep: fails at q in [b-ps-z]* Only 1 matches: polka	B. 1